Feasibility study outline — PPE and fall detection in shipyard work areas

Jinwoo Seol · personal example, November 2026 · independent work, not affiliated with any vendor or shipyard. Grey boxes show numbers from a public-data pilot, not from a shipyard.

1. Decision

Go / no-go for a paid pilot on **N fixed cameras** (e.g. block-assembly hall, dry dock, paint shop). Success criteria are written down before any data is seen: per-event recall at an agreed false-alert budget, alert latency, and hardware cost per camera.

Pilot evidence · Rehearsed on public data: SH17, 8,099 photos of people at work, 17 PPE and body-part classes, 922 photographers, with a fine-tuned YOLO26s. No shipyard data was used.

2. Data request

Per camera: view, height and distance; day / night / rain / welding glare / fog. Typical work and the rare events (falls, work at height, missing harness). A **class list with written label rules**, e.g. helmet = hard hat worn on the head (held in hand is not worn); fallen = torso on the ground for over 2 s (kneeling is not fallen). 2–4 weeks of footage, with the site's privacy (PDPA) handling agreed.

Pilot evidence · In SH17 a helmet is labelled on top of a head box, while gloves and bare hands are exclusive. 15 of the 16 'bare foot called shoes' errors were socked feet, which SH17 labels as foot: the rule book decides what counts as an error.

3. Site-condition test set

Held out **by camera and by date**, never by random frame. Report per-class precision and recall, and the look-alike pairs that matter: kneeling vs fallen, helmet vs cap / hood / bare head, gloves vs bare hands, harness vs vest straps.

Pilot evidence · Test: 1,628 photos from 307 photographers never seen in training. A random image split of the same size, where 92% of test photos share a photographer with training, scored 14.5 points higher mAP@0.5 (0.635 vs 0.490). The gain sits in rare classes (face-guard +47, medical-suit +41, ear-mufs +32); the 5 most frequent classes moved by at most 2.4 points.

4. Baselines

Prompted ZERO out of the box first, then fine-tuned on site data; report both on the same site test set, so the customer sees what the fine-tuning buys and what it costs in labelling.

Pilot evidence · Fine-tuned only: mAP@0.5 0.49, mAP@0.5:0.95 0.31 on held-out photographers. A zero-shot baseline was not run here.

5. Operating point

An **alert budget per camera per shift**, agreed with the safety team, because a missed fall and a false alarm cost different things. Thresholds are set per class from that budget, then confirmed over k of n frames before an alert is sent.

Pilot evidence · At ≤ 20 false alerts per 1,000 frames: bare hands recall 49% (precision 98%), helmet recall 46% (precision 72%). Of the 19 helmets detected on heads labelled bare, 12 were real hard hats nobody labelled, so a label audit moves the operating point. At one analysed frame every 10 s, an 8-hour shift is ~2,900 frames per camera.

6. Hardware and latency

Edge (Jetson Orin class) or on-premise server (L4 / A10 class); frames per second per stream, streams per GPU, alert latency end to end. The engine is built per GPU, so the container builds it on first start.

Pilot evidence · TensorRT FP16 on a laptop RTX 5070 Ti, 640 px: 1.0 ms model latency, 341 FPS per frame with resize and post-processing at batch 1, mAP@0.5 0.489 (PyTorch FP32 0.490). 1,426 images/s at batch 8: an upper bound of ~285 streams at 5 analysed FPS, before video decoding.

7. Risks and next steps

Data gaps (night, rain, real falls are rare) → generated data for rare hazards, but a **real-only** test set. Label drift between annotators → a written rule book and a double-labelled audit sample. Timeline: week 1 data and rules · weeks 2–3 baseline and fine-tune · week 4 site test and operating point · week 5 go / no-go review. From the site: camera access, sample footage, a safety-team contact.

Pilot evidence · Gaps of this pilot: no falls, no video, no night or rain conditions; stock photos, not CCTV.

Reproduce: `python run_all.py` rebuilds the split, model, metrics, TensorRT engines and this page. Data: SH17 (CC BY-NC-SA 4.0, Pexels photos). Code: AGPL-3.0, built on Ultralytics YOLO26s.

Pilot: recall against false alerts per 1,000 frames for the look-alike pairs (held-out photographers). The dot is the operating point.

